



Materials Chemistry



南方科技大学

Materials Chemistry Class

- Monday (单、双) 10:20-12:10, Teaching Building, Room 305
- Thursday (双) 8:00-9:50, Teaching Building, Room 305



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2022秋季学期材料化学

群号: 641288084



扫一扫二维码, 加入群聊。





Final scores:

Attendance and quiz: 15%

Assignments: 20%

Middle Exam: 25%

You will make an oral presentation on a specific topic to your class.

3 students to form a group and 30 min for each talk

Final Exam: 40%

Written examination



ETHICS: NO CHEATING AND PLAGIARISM



All these improper behaviors will cause you to fail the exam!

No excuses!



Outline

- Introduction to materials chemistry
- Solid State Chemistry
- Metals
- Semiconducting Materials
- Organic Soft Materials
- Nanomaterials and Nanotechnology



CHAPTER 1

What is Materials Chemistry?



Definition

Materials: Solid-state component or device that can be used to address a current or future societal need; it excludes complex liquid compounds such as crude oil, which may be more properly considered a precursor for materials.



Please Look Around



Classification

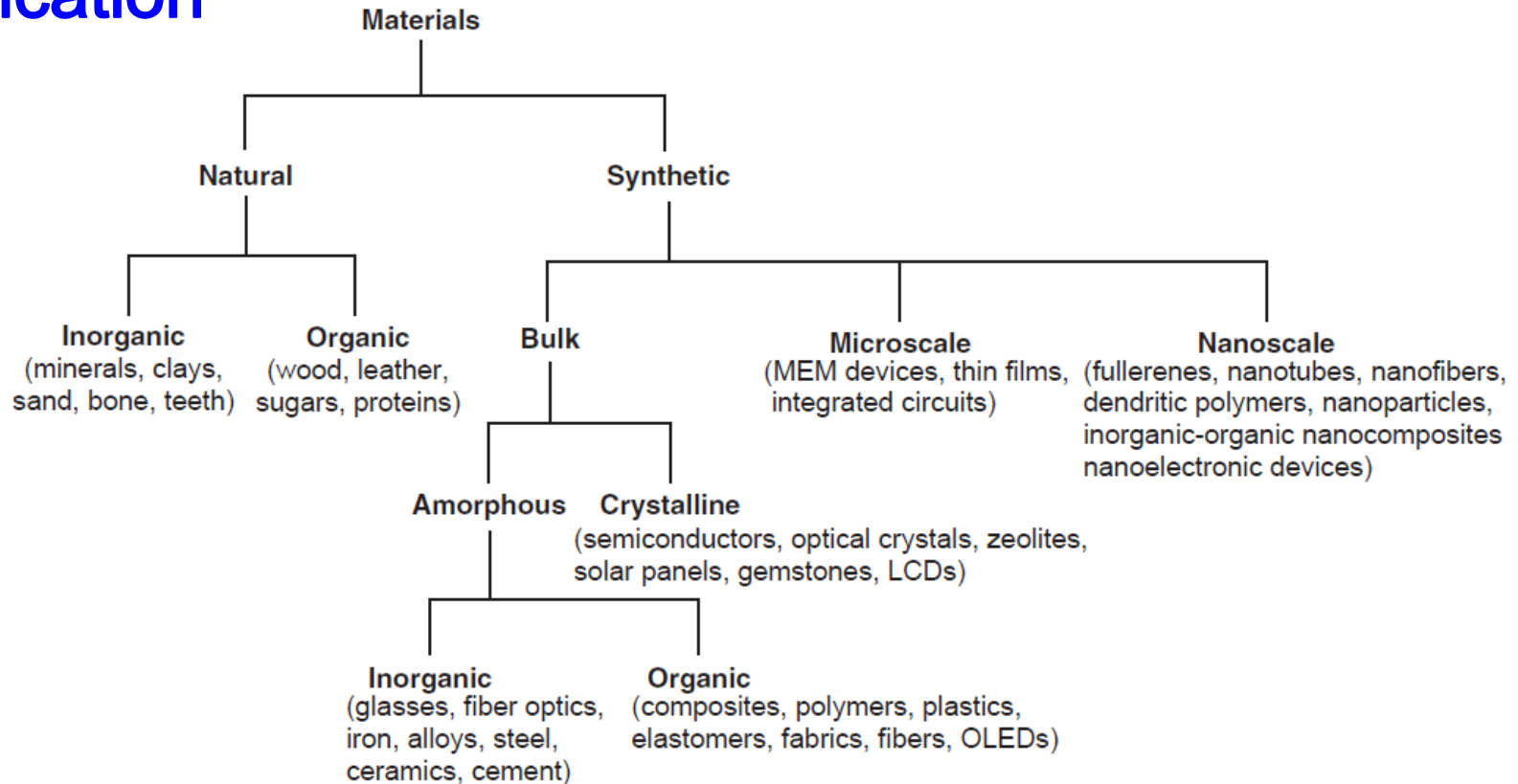


Figure 1.1. Classification scheme for the various types of materials.

Not exactly

e.g. Liquid crystals : between amorphous and crystalline phases
organic-inorganic hybrid composites



Classification

Metals: are materials that are normally combinations of "metallic elements". Metals usually are good conductors of heat and electricity. Also, they are quite strong but malleable and tend to have a lustrous look when polished.

Ceramics: A ceramic is an inorganic, nonmetallic solid material comprising metal, nonmetal or metalloid atoms primarily held in ionic and covalent bonds. The crystallinity of ceramic materials ranges from highly oriented to semi-crystalline, and often completely amorphous (e.g., glasses). Typically they are insulating and resistant to high temperatures and harsh environments.

Classification

Plastics: (or polymers) are generally organic compounds based upon carbon and hydrogen. They are very large molecular structures. Usually they are low density and are not stable at high temperatures.

Composites: consist of more than one material type. Fiberglass, a combination of glass and a polymer, is an example. Concrete and plywood are other familiar composites. Many new combinations include ceramic fibers in metal or polymer matrix.



Classification

Newer Branches

Semiconductors: have electrical properties intermediate between metallic conductors and ceramic insulators. Also, the electrical properties are strongly dependent upon small amounts of impurities.

Nanomaterials: Nanomaterials are defined as materials with at least one external dimension in the size range from approximately 1-100 nanometers. Example - miniature robots to do surgery inside the body or miniature chemical laboratories and instruments that will continuously analyze blood and dispense medications inside the body.

Discipline

Materials Chemistry is focused on understanding the relationships between the arrangement of atoms, ions, or molecules comprising a material, and its overall bulk structural/physical properties.

Goal:

Discovery and design of new materials.



1.1 Historical perspectives

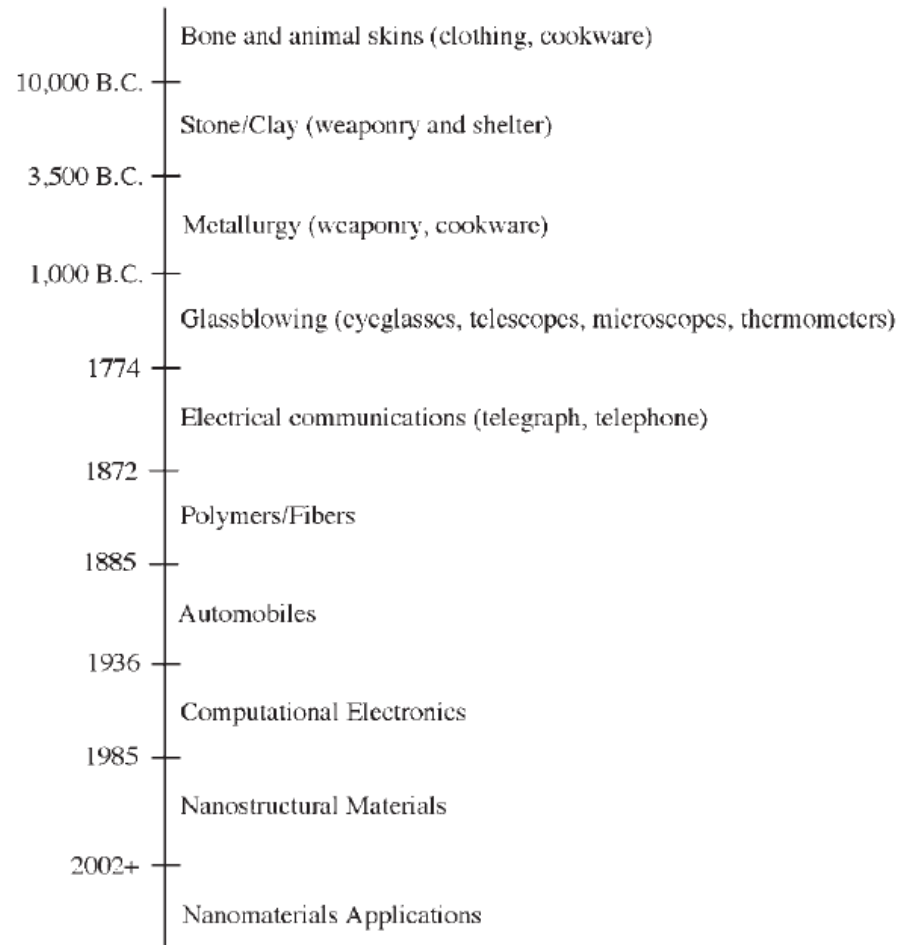


Figure 1.2. Timeline of major developmental efforts related to materials science.



- Neolithic Age (10,000-300 B.C.)



Limestone, wood, shells and clay
Shaping ↔ Functions

- ✓ Utensils
- ✓ Tools
- ✓ Weaponry





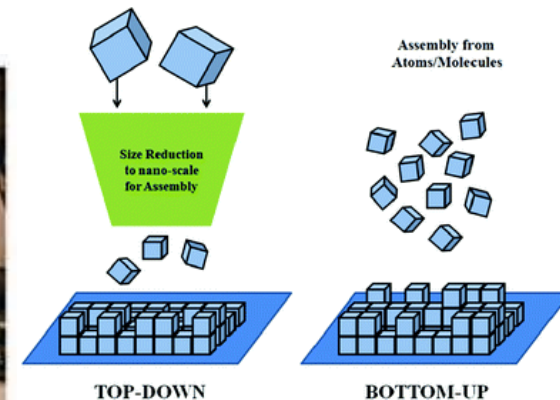
- Chalcolithic Age (4,000-1500 B.C.)



Copper

From oxide ores (malachite)
“top-down” materials synthesis

Realize materials properties:
malleability and thermal conductivity



- **Bronze Age (1,400-0 B.C.)**



Metal Alloys

Arsenic-doped copper Lautite and domeykite ores

Tin-copper alloys (bronze)

- ✓ Lower melting point
- ✓ Higher hardness
- ✓ Lower brittleness



- Iron Age (1,000-1,950 B.C.)



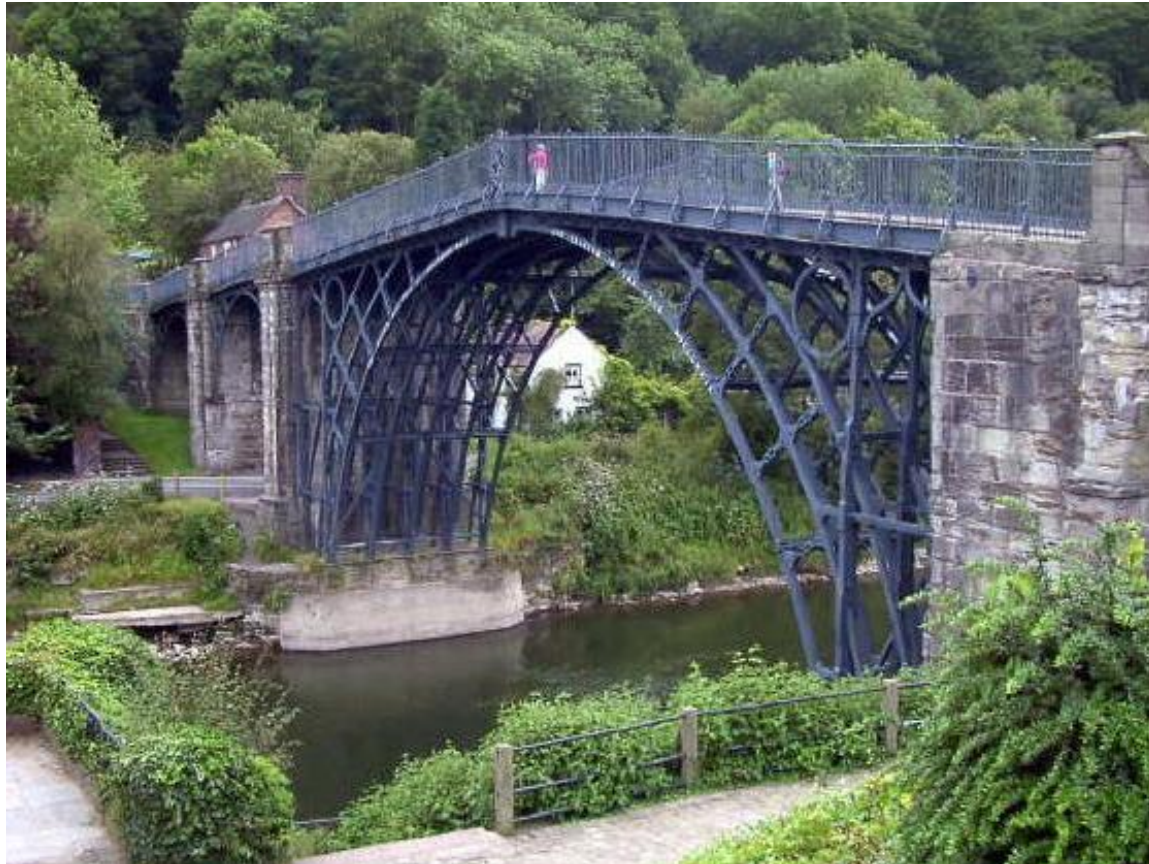
**Abundant
in the earth's crust**

Wrought iron

Steel (< 2%)

Cast iron
Iron-carbon (> 2%) alloys
Chinese





Ironbridge in Shropshire

The world's first cast iron bridge was built over the River Severn at Coal Brookvale in 1779.



Golden Gate Bridge

1937: steel became available. This had increased strength and lower brittleness. This allowed larger structures to be made.



Shenzhen Bay Bridge

2003: 5545 m



1.2 Considerations in the design of new materials

➤ Current societal need

➤ Availability of resources

New energy resources

➤ Cost

e.g. Ge-based transistors \$8-10

Si-based transistors \$one-millionth of a penny

Life is changing--- more controllable

Top-down

The transformation of complex natural products into desirable materials.

Bottom-up

Traditionally

Combining/molding bulk precursor compounds

Nanotechnology

The design of materials from the atomic level;

Providing an unprecedented control over resultant properties.



Molecular machine

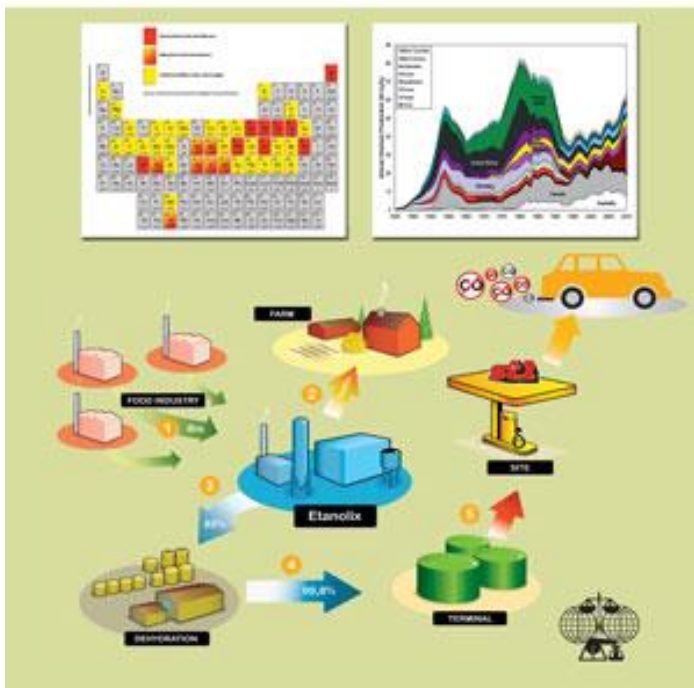
Nanotechnology enables new materials to be synthesised at a molecular and atomic level.



Sustainable materials

Materials for a Sustainable Future

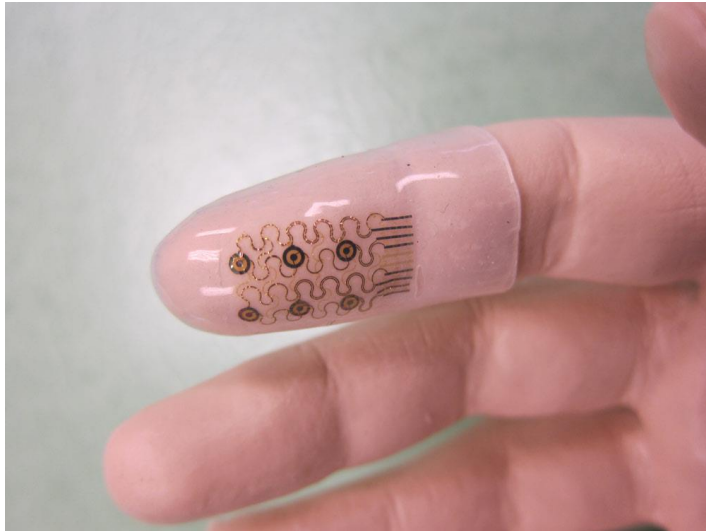
Edited by Trevor M Letcher and Janet L Scott



- ✓ bioplastics;
- ✓ water purification,
- ✓ water-splitting,
- ✓ photovoltaic cells,
- ✓ batteries and fuel cells;

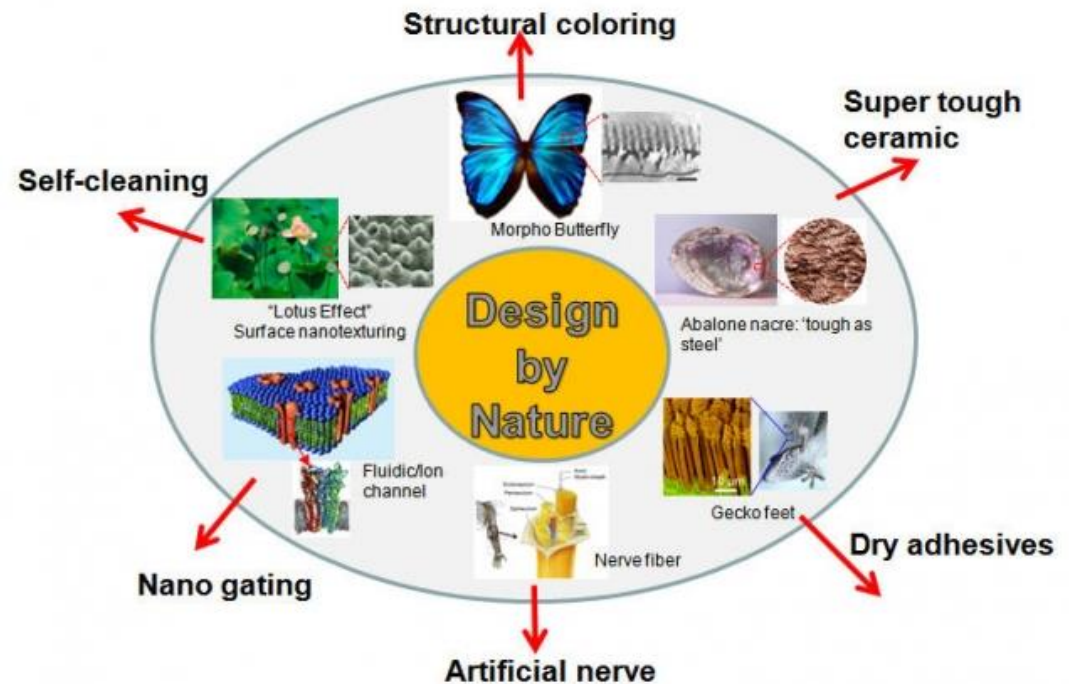


Smart materials



Flexible electronics

Bio-inspired materials





Nanomedicine

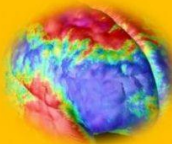
Diagnostics

In vitro
& In vivo



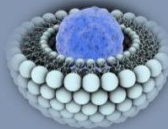
Medical Imaging

In vivo



Nano-therapeutics

Systems &
Devices

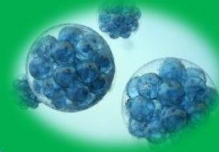


Vaccines



Regenerative Medicine

Biomaterials
Functionalisation





1.3 How to design a new material?

Critical thinking

Literatures:

Nature; Science;

The Journal of the American Chemical Society;

The Chemistry of Materials

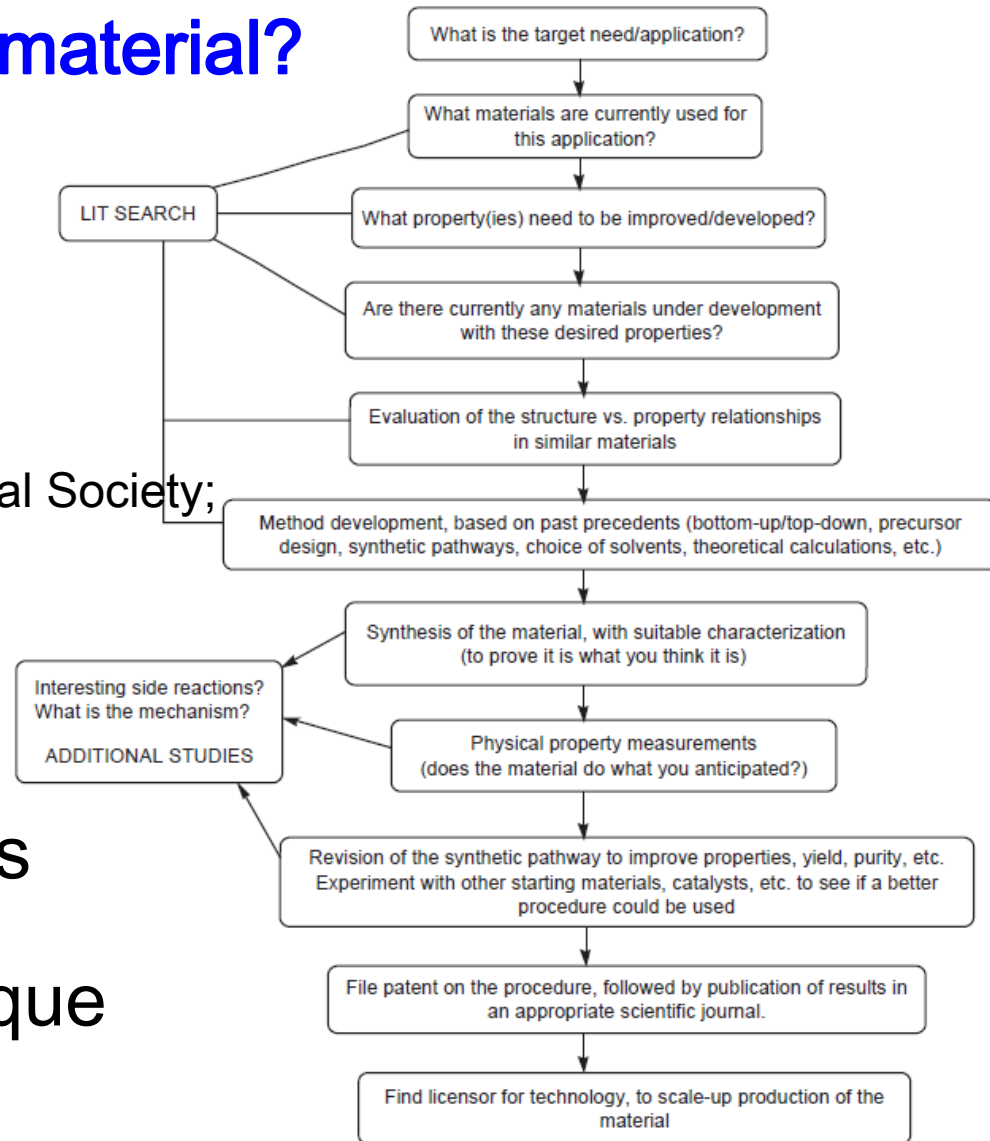
Advanced Materials;

Nano Letters;

Small etc.

Synthetic methodologies

Characterization technique





Homework (1st)

1. Pick a kind of material you are interested in.
2. Search and download a related literature (most cited or latest) from the noted material journals.
3. According to the flow chart on PP26 to condense the literature to a short essay (300-500 words).
4. Application; recent progress (developed and need to be improved); preparation methods; structure-properties relationship; the significance; prospect.

Deadline: Sep. 12; Please send a hard copy of your essay.
LATE HOMEWORK WILL NOT BE ACCEPTED !
ZERO TOLERANCE FOR PLAGIARISM!